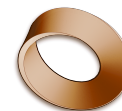


Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics



1.3 – Refraction

Unit 1: Light & the Electromagnetic Spectrum

Learning intentions

- a) I can explain why light bends when it passes from one transparent material into another.
- b) I can describe how the direction of bending depends on whether light speeds up or slows down.
- c) I can predict the path of a light ray passing through a rectangular block based on the angle of incidence.

Theory

When light travels from air into a denser transparent material, such as **glass** or **perspex**, its speed changes. This change in speed causes the light ray to change direction, a phenomenon known as **refraction**. If the light enters the glass at an angle (not along the normal), it slows down and bends _____ to the normal line. The normal is an imaginary line drawn at 90° to the surface of the material where the light hits.

Conversely, when light travels from a denser material like glass back into air, it speeds up again. This increase in speed causes the ray to bend _____ from the normal as it exits the block. It is important to remember that refraction only occurs if the light hits the boundary at an angle. If the light ray hits the surface exactly along the normal (at 90° to the surface), it passes straight through without bending, even though its speed still changes.

To investigate this, we often use a rectangular glass or perspex block and a ray box. By tracing the path of the light ray entering and exiting the block, we can observe how the entry angle affects the exit angle. The amount of bending depends on the **refractive index** of the material, which tells us how much slower light travels in that material compared to air.

Word bank: away · towards · straight · faster · slower

Activity

Investigating Refraction in a Rectangular Block

You will need a rectangular glass or perspex block, a ray box with a single slit, a piece of white paper, a pencil, and a ruler.

- Place the rectangular block in the centre of your paper and draw its outline carefully with a pencil.
- Shine a narrow beam of light from the ray box so that it hits one side of the block at an angle (not 90°). Mark two points on the incident ray (before it enters) and two points on the refracted ray inside the block, and two points on the emergent ray (after it leaves).
- Remove the block and draw straight lines through your marked points to complete the path of the light ray.
- Draw a normal line (dashed) at 90° where the light enters the block. Measure the angle between the incident ray and the normal (angle of incidence) and the angle between the refracted ray inside the glass and the normal (angle of refraction).
- Repeat steps 2–4 for three different angles of incidence (e.g., 30° , 45° , and 60°).

Record your results in the table below:

Angle of Incidence (°)	Angle of Refraction (°)	Did the light bend towards or away from the normal?
30	_____	_____
45	_____	_____
60	_____	_____

Questions

1. Explain why the light ray bends when it enters the glass block from the air.

2. If the angle of incidence is increased, what happens to the angle of refraction?

3. Why does the emergent ray (leaving the block) appear parallel to the original incident ray, even though it has been displaced sideways?

4. A student claims that light bends because it "wants" to take the shortest path. Using the concept of speed change, explain why this explanation is incomplete and what actually causes the bending.

Extension

Research the term **total internal reflection**. Under what specific conditions does light fail to exit a glass block and instead reflect back inside?
